

A study report: architecture, physics library, optimizer (V1/V2/V3), backends, and honest limits

What it is

You give it a MISSION (agility, speed, endurance, sensing) and it produces a BUILDABLE vehicle — geometry, a subsystem architecture, a physics model, and the flight-controller config — plus a three-tier optimizer that can tune it, rearrange it, and (in prototype) invent past it. Every number is the real coupled physics solve; nothing is a fitted proxy. The engine is domain-general; the UAV is instance one and the domain we can check.

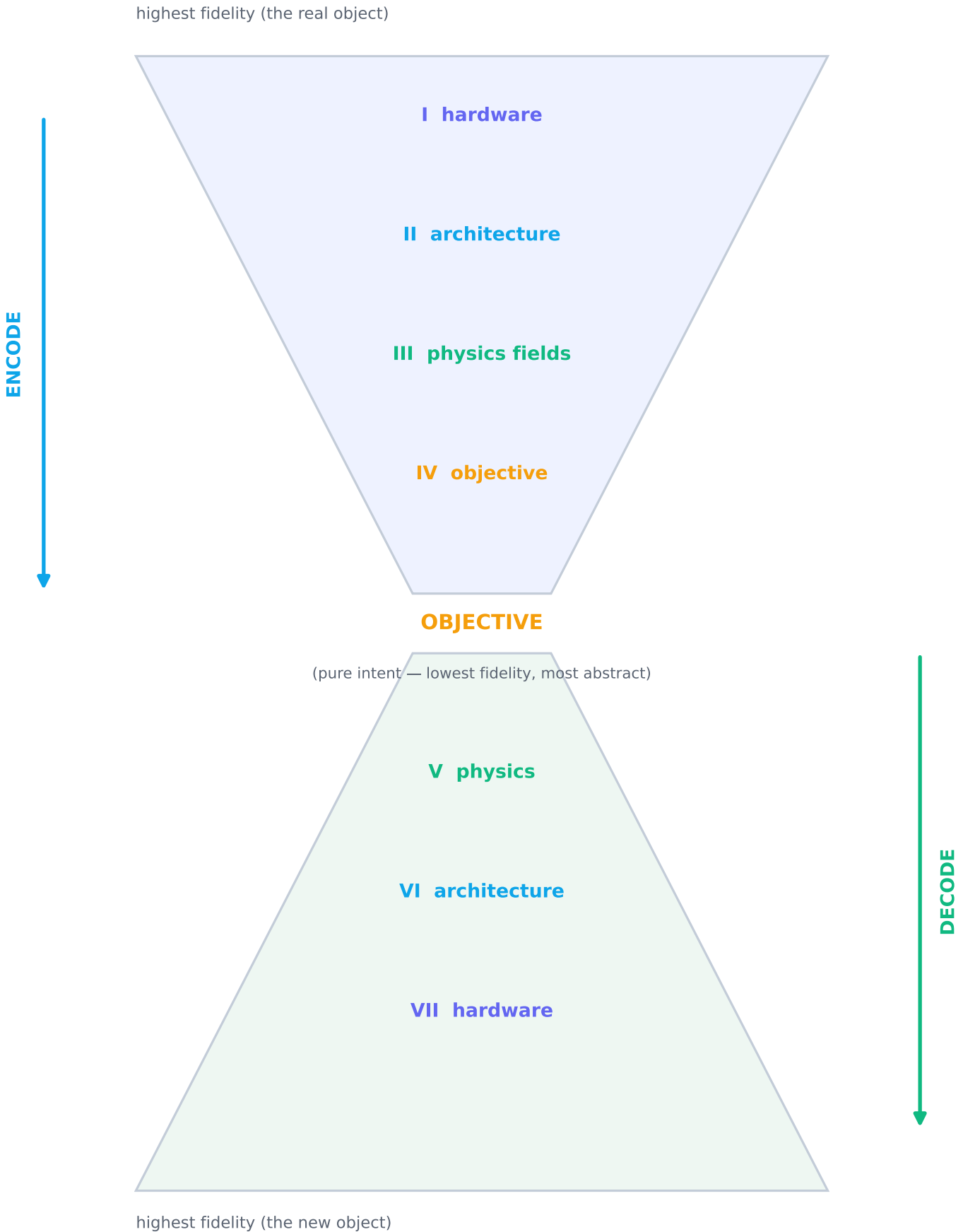
End-to-end result — counter-UAS interceptor (headless, this run)

mission: interception · score: max intercept hits · validation: PASS (8/8 checks)	
platform	6-rotor · electric · propeller · ArduPilot/CubeOrange
capabilities	a_max 6.64 g · v_max 100 m/s · endurance 59 min
seeker (EO/IR)	detect 2244 m · FOV 18.3° · track 104 Hz
SCORE	MAX INTERCEPT HITS = 7 / 12 (from baseline 1)
package	STL · STEP · OpenFOAM CFD · CalculiX/gmsh FEA · .param · arducopter.apj

The honest thesis

The pipeline reaches 7/12 and every airframe tier plateaus there — because the binding limit is the SEEKER, not thrust. That finding, and the clear separation of what is validated vs prototype vs frontier, is the point of this report: an engine that states what it grounds AND what it can't.

ENCODE abstracts a concrete vehicle down to intent; DECODE regenerates a vehicle from intent



Two worlds: the DARK (abstraction / law space — infinite, where it imagines) and the LIGHT (the coupled solve — definite, where reality says yes or no). V3 is the traffic between them.

The Physics Sanctuary — the library, and how to leave it

Grounding to a curated law graph; and deriving law FORMS from first principles instead

PHYSICS LIBRARY (semantic memory)
~1000+ law nodes, grounded, curated

each node: quantity · law · requires · points_to
+ assumptions (validity envelope)

Descent to fundamentals = a validation tree:

rotor_thrust

actuator_disk

conservation_of_momentum

navier_stokes

If the ROOTS are validated and each derivation edge is sound, every node inherits validity.

Decoupling: physics WITHOUT the library (V3d)

DIMENSIONAL KERNEL
(Buckingham-Pi)

resolved FIELD
(CFD / FEM)

complete LAW
(no library)

FORM from units alone: $\text{drag} = C \cdot \rho \cdot v^2 \cdot A$ ($\text{thrust} = C \cdot \rho \cdot D^4 \cdot \Omega^2$)

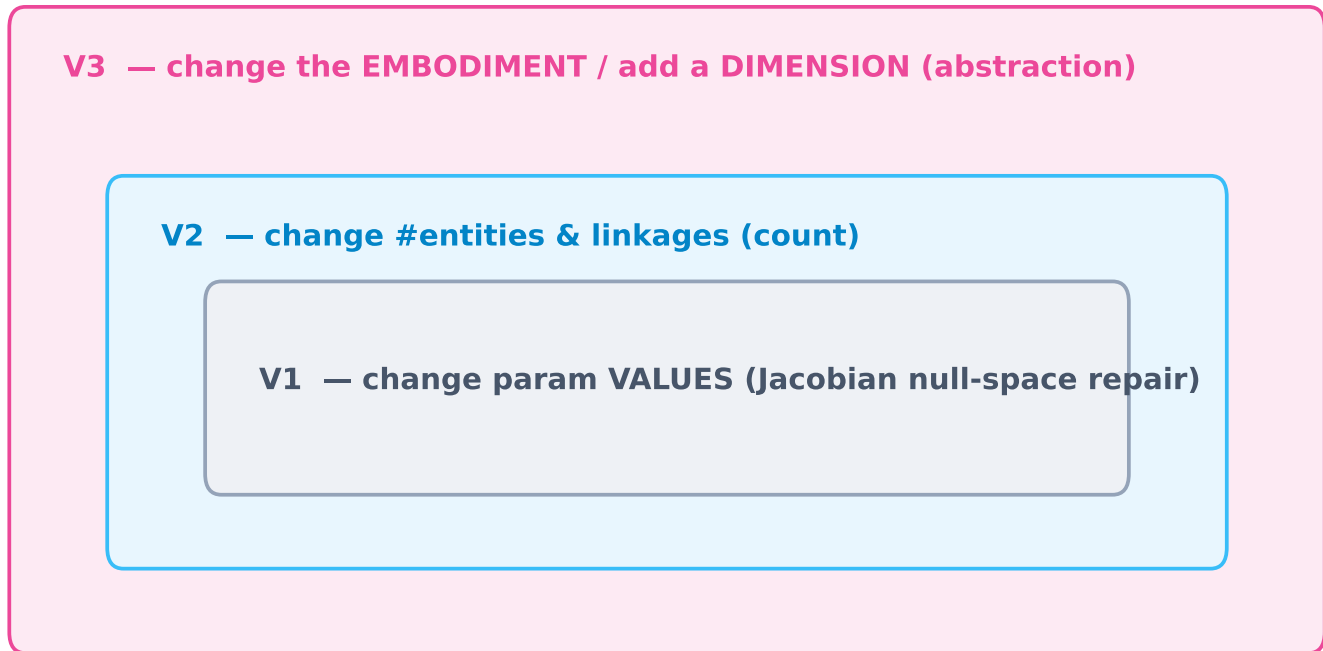
CONSTANT from the solve: $C_d = 0.48$ (CFD, 27k cells) $\rightarrow \text{drag} = 0.2414 \cdot \rho \cdot v^2 \cdot A$

$\rightarrow$ library demotes to a CACHE of the few empirical constants; the forms are generated.

Honest boundary

- Dimensional analysis gives the FORM/scaling, never the constant C or $f(\text{Re}, \dots)$.
- The constant comes from a resolved field — which itself rests on measured MATERIAL constants
- (viscosity, moduli): the residue descends, it does not vanish.
- A full law is $C(\text{Re})$: a SWEEP of solves; and only an exact (DNS) solve is purely first-principles.

A nested ladder of design freedom; each frees what the one below holds fixed



The four V3 modes (and their motivation):

V3a compositional

motivation: select an existing embodiment from the library

result: collapses to V2 (only realizes registered models)

V3b generative

motivation: dissolve to a field, reshape it, RE-EMBODY a new form

result: produces the ducted ring — NEW form, but UNVALIDATED

V3c meta-requirement

motivation: over-constrained field → prescribe a MISSING DOF

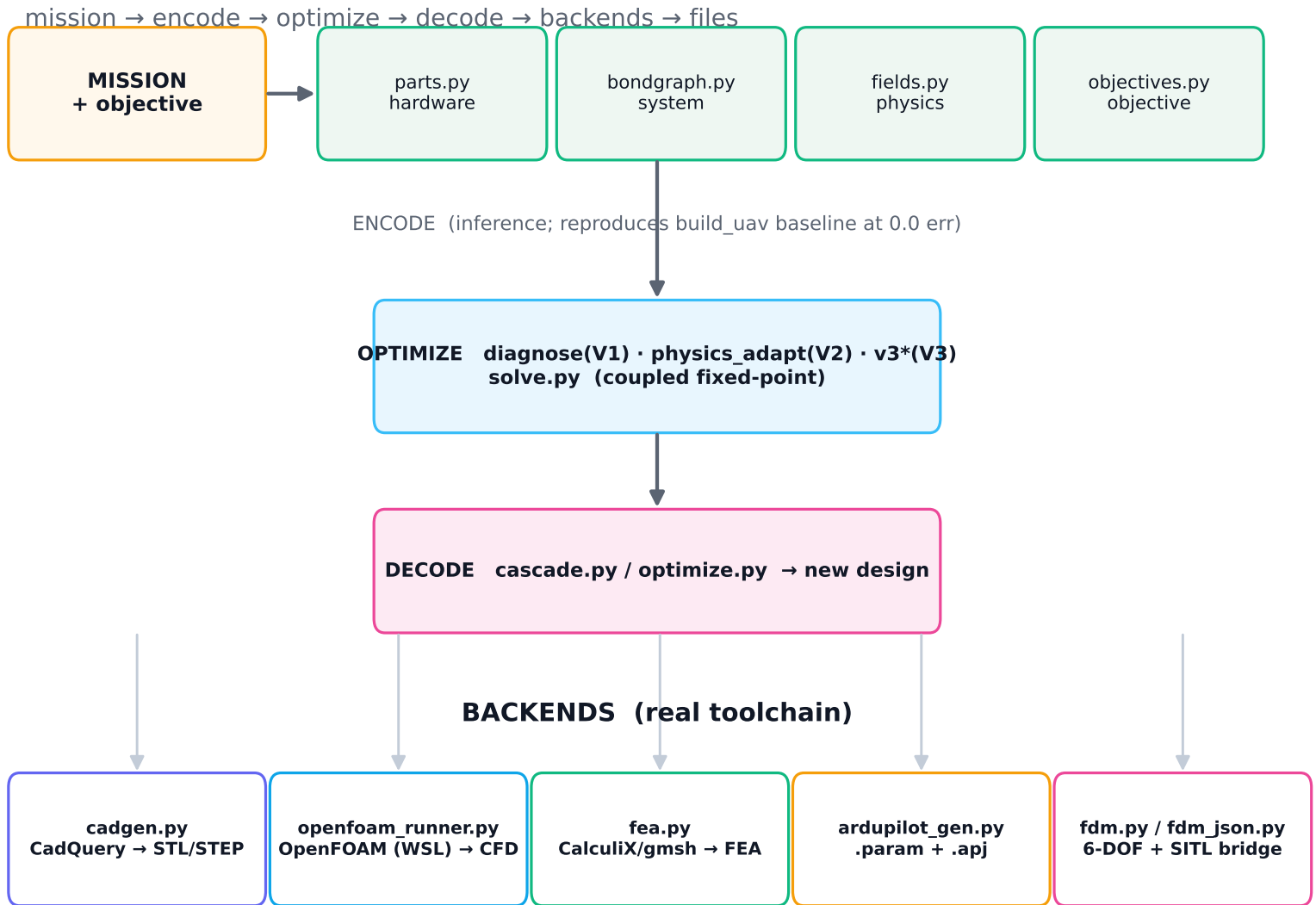
result: 'seeker needs a scan DOF' — decoupled from the gimbal

V3d decoupling

motivation: derive the law FORM from units + constant from a solve

result: a complete library-free drag law (proven)

Escalation: V1 fires first; on exhaustion (null space collapses) V2; then V3. One metric— radicality — measures how far e



Deficit-driven dispatch:

a field inside its reduced-model validity uses the fast model; when it leaves validity (tip-Mach, disk-loading), it is handed to the external solver — the leash is a proxy for validation.

one command (validate_pipeline.py) → a complete buildable package

build_interceptor/

hardware/ specimen.step, specimen.stl

CAD B-rep + mesh

cfdf/of_flow_case/

OpenFOAM case: system/ constant/polyMesh (27k cells) 0

system/*Dict, constant/turbulenceProperties, postProcessing/forceCoeffs

fea/ arm.inp, arm.msh, arm_result.json

CalculiX deck + gmsh mesh + solved beam FE

ardupilot/ specimen.param

41 generated params (frame, battery, ESC, seeker mount,

ardupilot/ arducopter.apj (1.50 MB)

official CubeOrange firmware (downloaded, sha256, prove

ardupilot/ firmware_manifest.json

board_id, git id, source URL, checksum

MANIFEST.json

design + capabilities + per-artifact provenance

Provenance is tagged per artifact — nothing fabricated:

STL/STEP (CadQuery), .param (from the design), FEA deck+mesh

CFD drag (OpenFOAM RANS), beam-FE stress

official (real) ArduCopter firmware, not tool-authored — cannot compile here)

same mission; what each tier changes, what it returns, its cost & status

tier	what it changes	mechanism	result (a≥5g,v≥26,e≥16)	status
baseline	—	—	UNMET (2.54 g)	—
V1	param values	Jacobian null-space repair	MET · 3.49 kg (4 rotors)	real
V2	+ rotor count / linkage	discrete search wrapping V1	MET · 3.04 kg (3 rotors)	real
V3a compositional	+ select embodiment	ascend→imagine→select (library)	= V2 (collapses)	real
V3b generative	+ generate embodiment	reshape field → re-embody (ring)	8.4 g/73 min/2.11 kg	PROTOTYPE
V3c meta-req	+ prescribe a missing DOF	over-constraint → étendue	seeker: 'add scan DOF'	real (hint)
V3d decoupling	+ derive the physics	dim. kernel + field solve	drag = 0.24·pv²A (no lib)	real (drag)

Reading it:

- V1 does the real lifting; V2 beats it (lighter, feasible) by adding rotor count on top of tuning.
- V3a folds into V2 — pure selection isn't new. V3b/c/d are where V3 exceeds V2 (new form / new dimension / derived physics), but V3b is an unvalidated model and V3c is a hint, not a built part.
- On the interception score every airframe tier plateaus at 7/12 — the wall is the SEEKER, not thrust.

the line between validated, prototype, and frontier

CAN (real & verified)

- ✓ mission → buildable package, end to end (STL/CFD/FEA/.param/.apj), 8/8 validation
- ✓ V1 params + V2 count: genuinely different, V2-dominant results
- ✓ encode inference reproduces the hand-built baseline exactly (0.0 err)
- ✓ reduced physics for 6 disciplines incl. a modeled EO/IR seeker
- ✓ CFD (OpenFOAM/WSL) for airframe drag; FEA deck + solved beam; 6-DOF FDM + SITL bridge
- ✓ V3: imagine alternatives, reshape a 2-D field, hint a missing DOF, derive law FORMS from units
- ✓ V3d: a complete library-free drag law (form from units + constant from CFD)

CAN'T / DOESN'T (yet) — flagged, not hidden

- ✗ validate V3's invented embodiments — the ducted ring is an UNVALIDATED deficit model
- ✗ rotor-thrust CFD (only body drag); solve arm.inp (no ccx → 1-D beam); compile the .apj (no ARM toolchain)
- ✗ fly live SITL (bridge verified in loopback only); fold CFD drag back into the caps
- ✗ V3 RESCUE a mission — it folds into V2 unless it invents, and inventions aren't trusted
- ✗ choose its own objective; invent unencoded physics / embodiments; C(Re) sweeps
- ✗ other platforms/powertrains, and disciplines thermal / RF / radar / guidance-miss / wind

The one honest throughline

Imagination is free; realization is the wall. V3 can abstract, reshape, and derive — but a form is only trustworthy when a validated model or solve backs it. Validation propagates UP a sound derivation, so validating the small physics validates the higher — the leash lengthens exactly as far as the derivation stays sound and in-envelope. That boundary is the real remaining work.